

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Botany**Paper: BOB-501 — Comparative Studies of Cryptogams**

Time: Three hours

Max. Marks: 70

Note: Attempt five questions in all, selecting one question each from Section-A, B, C, and D. Section-E is compulsory. The figures in the right-hand margin indicate marks.

SECTION – A

Question 1. Give an illustrated account of sexual reproduction and post-fertilization development in Florideophyceae. [15]

Question 2. Write notes on any two of the following: [7.5 × 2 = 15]

- (a) Pigmentation in algae
- (b) Life history types in Phaeophyceae
- (c) Volvocine line of evolution
- (d) Methods of asexual reproduction in Cyanophyceae

SECTION – B

Question 3. Give an account of Ainsworth's system of classification of fungi. Also mention the salient features of different classes. [15]

Question 4. Write notes on any two of the following: [7.5 × 2 = 15]

- (a) Types of fruiting bodies in Ascomycotina
- (b) Types of fungal spores
- (c) Economic importance of fungi
- (d) Types of life-cycles in fungi

SECTION – C

Question 5. Give an illustrated account of evolution of sporophytes in bryophytes by progressive sterilization of sporogenous tissues. [15]

Question 6. Write critical notes on any two of the following: [7.5 × 2 = 15]

- (a) Classification of bryophytes (Smith's System)
- (b) Affinities of bryophytes with algae

- (c) Vegetative and asexual reproduction in bryophytes
- (d) Range of gametophyte structure in thalloid bryophytes

SECTION – D

Question 7. Describe the evolution of stelar system in pteridophytes. [15]

Question 8. Write comprehensive notes on any two of the following: [7.5 × 2 = 15]

- (a) *Marsilea* sporocarp
- (b) *Psilotum* sporophyte
- (c) Types of sori in ferns
- (d) Morphology of *Ophioglossum* sporophyte

SECTION – E

Question 9. Differentiate between and answer the following:

(a) Differentiate between any five of the following: [1 × 5 = 5]

- (a) Unilocular and plurilocular sporangia
- (b) Trichoblast and gonimoblast
- (c) Smut and rust fungi
- (d) Pseudoelaters and elaters
- (e) Leptosporangiate and eusporangiate ferns

(b) Give very brief answers for the following (in not more than 100 words each): [1 × 5 = 5]

- (a) Name the food reserves present in Cyanophyceae.
- (b) Function of heterocyst.
- (c) Name two edible fungi.
- (d) Write the name of a pteridophyte which can be used as biofertilizer.
- (e) Write the ecological importance of *Sphagnum*.

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